

Validation targets support different claims

Reference / comparison	What it can support	Not established alone
Anatomy / image similarity	Consistency with segmentation or an image-matching criterion	Dense material correspondence or accurate strain gradients
Inter-software agreement	Consistency of specified outputs between tested implementations	Material-motion accuracy or absence of shared bias
Sparse landmarks	Correspondence near selected reference points	Accuracy between landmarks or throughout the strain field
Prescribed motion	Technical recovery within the simulated conditions	In-vivo accuracy or performance beyond the modelled regime
Tagging / DENSE	Agreement with a registered material-sensitive reference	Error-free truth or transfer to untested conditions
Scan-rescan	Precision under the stated repeat conditions	Accuracy or absence of systematic bias
LGE-defined tissue regions	Association with independently defined tissue regions	Dense motion truth or one-to-one tissue-strain correspondence
Clinical outcomes	Association or decision-related performance as directly studied	Treatment benefit without an appropriate evaluation design